# Corner detection: the math

How are $\lambda_{max}$, $x_{max}$, $\lambda_{min}$, and $x_{min}$ relevant for feature detection?

- What's our feature scoring function?

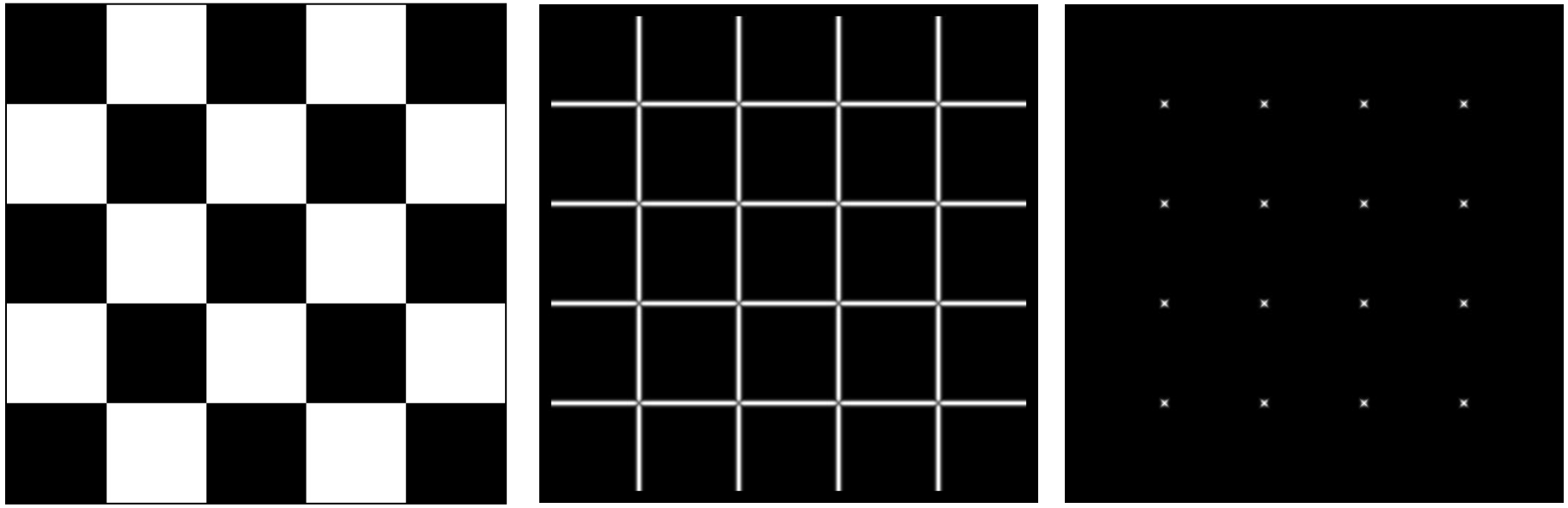
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Want $E(u,v)$ to be large for small shifts in all directions

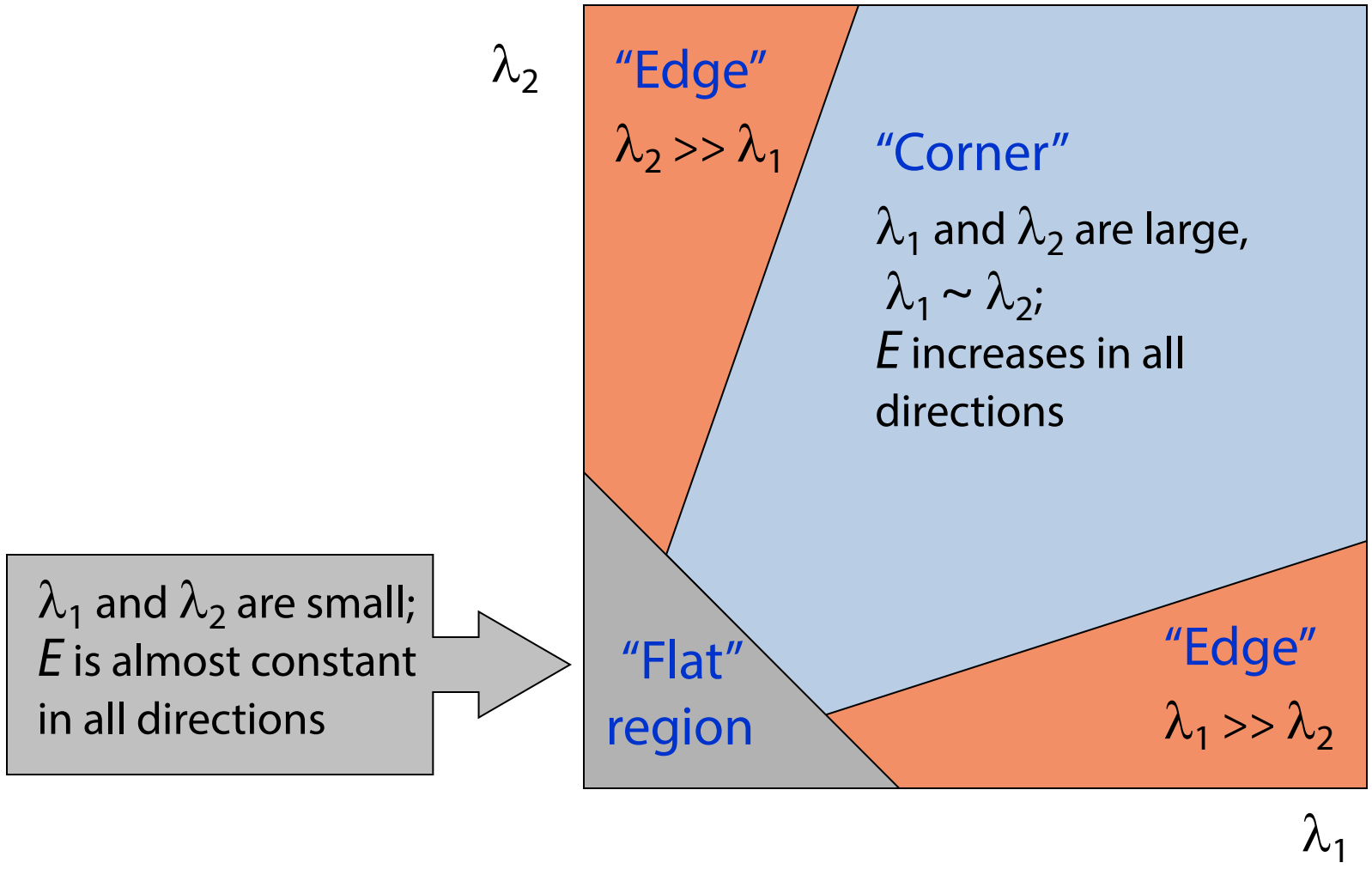
- the minimum of $E(u,v)$ should be large, over all unit vectors $[u\ v]$
- this minimum is given by the smaller eigenvalue ($\lambda_{min}$) of $H$



$I$ $\lambda_{max}$ $\lambda_{min}$

# Interpreting the eigenvalues

Classification of image points using eigenvalues of *M*:

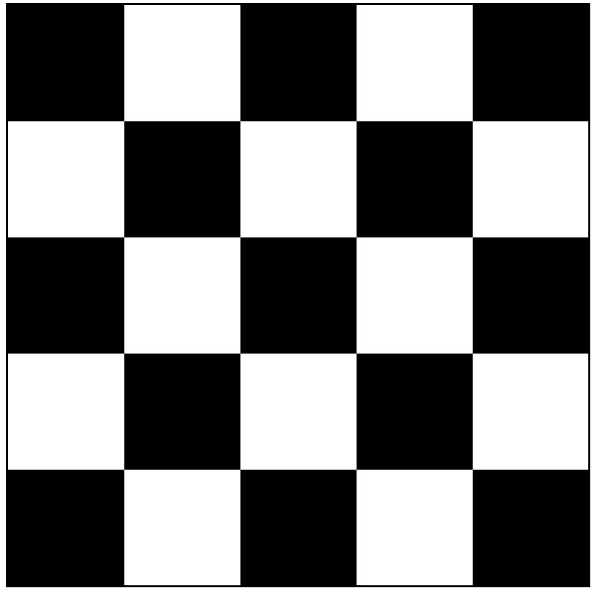


# Corner detection summary

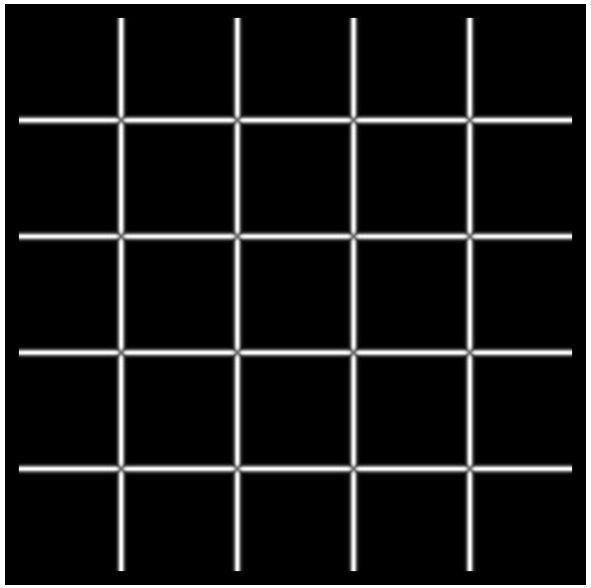
**Here's what you do:**

- Compute the gradient at each point in the image
- For each pixel:
  - Create the *H* matrix from nearby gradient values
  - Compute the eigenvalues.
  - Find points with large response ($\lambda_{min}$ > threshold)
- Choose those points where $\lambda_{min}$ is a local maximum as features

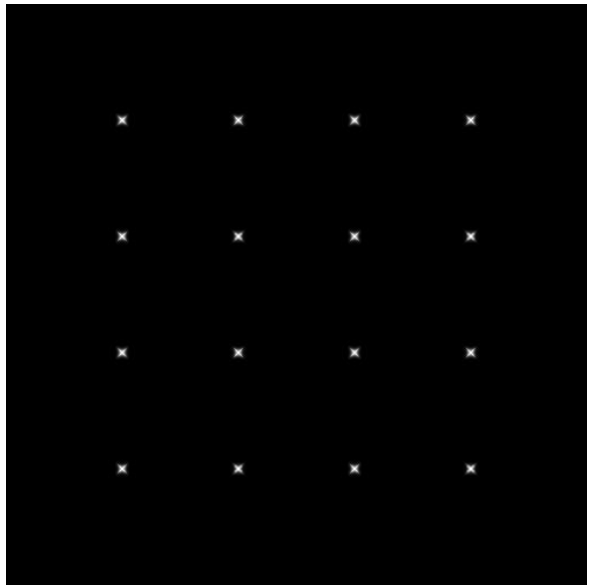
$$H = \sum_{(x,y)\in W} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$



$I$



$\lambda_{\max}$

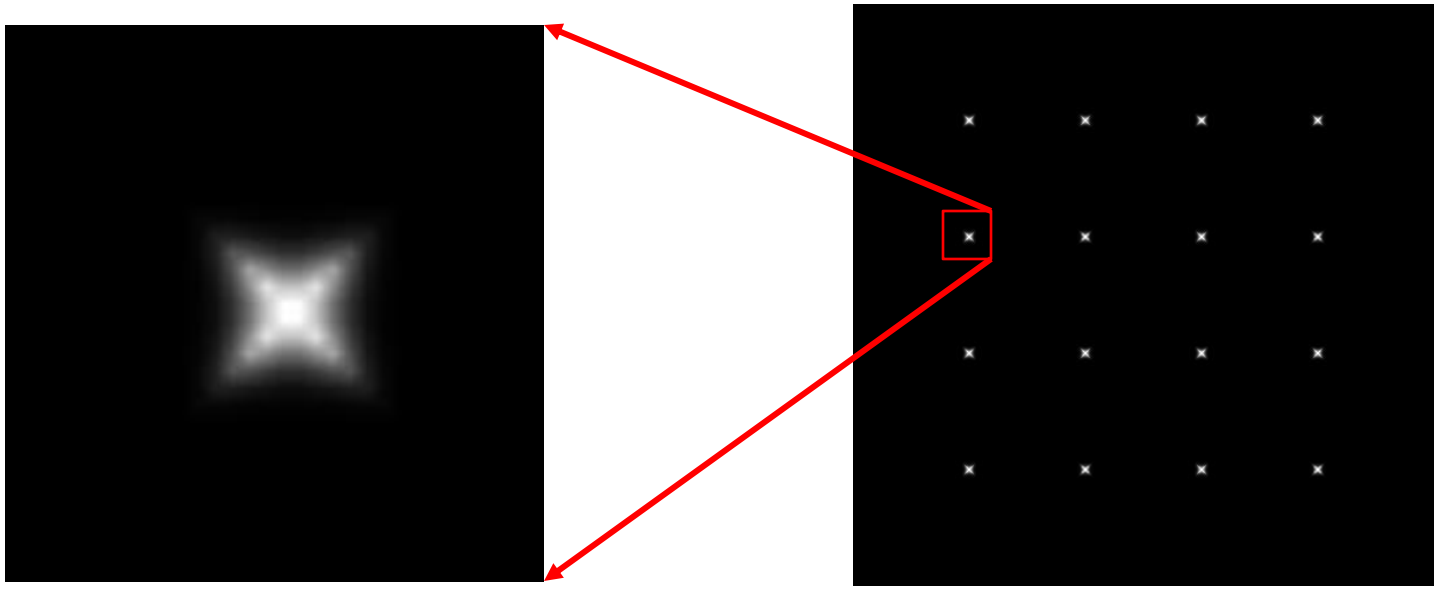


$\lambda_{\min}$

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# The Harris operator

$\lambda_{min}$ is a variant of the "Harris operator" for feature detection

$$f = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

$$= \frac{determinant(H)}{trace(H)}$$

- The *trace* is the sum of the diagonals, i.e., $trace(H) = h_{11} + h_{22}$
- Very similar to $\lambda_{min}$ but less expensive (no square root)
- Called the *Harris Corner Detector* or *Harris Operator*
- Lots of other detectors, this is one of the most popular

# Alternate Harris operator

- will use an alternate definition of the Harris operator:

$$R = \lambda_1 \lambda_2 - k \cdot (\lambda_1 + \lambda_2)^2 = \det(M) - k \cdot \mathrm{tr}(M)^2$$

# The Harris operator

